

F72 — Gindikin pin, the last factor's mechanism: the $15/128 = N_c \cdot n_C / 2^g$ factor splits into TWO substrate-architectural operations. $\times(N_c \cdot n_C)$: the Born rule ($|\cdot|^2$, T754) squares the conformal-group dimension $\dim SO(4,2) = N_c \cdot n_C$ (Euclidean carries it linearly via the kernel K ; the FK Born-rule measure carries it squared via $|K|^2$). $\div 2^g$: automorphism-invariance sheds the flat Euclidean-volume artifact (extensive $\rightarrow$ intensive). Structure grounded in T754 + F66; exact identification ($= \dim SO(4,2)$, $= 2^g$) is the multi-week FK normalization.

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F72 — The Born-squaring mechanism for the $15/128$ factor

0. The last open factor

F71 reduced $c_{FK} = 225/\pi^{9/2}$ to $c_{FK} = K(0,0) \cdot (N_c \cdot n_C / 2^g) \cdot \sqrt{\pi}$, with $K(0,0) = 2^g \cdot N_c \cdot n_C / \pi^{n_C}$ DERIVED and $\sqrt{\pi}$ the dimension-parity half-integer (Grace, resolved). The one remaining factor is $15/128 = N_c \cdot n_C / 2^g$. This note gives its mechanism — structurally, grounded in established results; the exact constants re-

main the multi-week FK computation.

1. The factor splits into two operations

$$\frac{N_c n_C}{2^g} = \underbrace{(N_c n_C)}_{\text{Born } |\cdot|^2 \text{ squares the conformal-group dim}} \times \underbrace{\frac{1}{2^g}}_{\text{auto-invariance sheds the flat Euclidean } 2^g}$$

Comparing the two measures (the dual- ρ pair): - Euclidean / Bergman: coefficient $2^g \cdot N_c \cdot n_C$ — $N_c \cdot n_C$ linear, times the flat 2^g . - FK Born-rule invariant: coefficient $(N_c \cdot n_C)^2$ — $N_c \cdot n_C$ squared, flat 2^g shed.

2. Why $\times(N_c \cdot n_C)$: the Born rule squares the conformal-group dimension

T754 (RIGOROUSLY CLOSED) defines c_{FK} as the Born-rule automorphism-invariant measure. The Born rule is probability = $|\text{amplitude}|^2$. In the Bergman/coherent-state picture, the amplitude between coherent states is the reproducing kernel:

$$\langle z|w \rangle = \frac{K(z, w)}{\sqrt{K(z, z)K(w, w)}}, \quad \text{Born density } \propto |\langle z|w \rangle|^2 = \frac{|K(z, w)|^2}{K(z, z)K(w, w)}.$$

So the Euclidean volume normalizes the kernel K (linear in K), while the Born-rule measure normalizes $|K|^2$ (squared). The “count” that appears linearly in K and gets squared by $|\cdot|^2$ is the dimension governing the kernel’s symmetry — and the Bergman kernel of D_{IV}^5 is covariant under $\text{Aut}(D) = \text{SO}(5,2)$, with the coherent states organized by the conformal subgroup $\text{SO}(4,2)$ (Casey #14, the F66 EM-boundary sector). Its dimension is $\dim \text{SO}(4,2) = N_c \cdot n_C = 15$. So:

Euclidean ($\propto K$): coefficient linear in $\dim \text{SO}(4,2) = N_c \cdot n_C$. Born-rule ($\propto |K|^2$): coefficient $(\dim \text{SO}(4,2))^2 = (N_c \cdot n_C)^2 = 225$.

The Born $|\cdot|^2$ is literally the squaring; the conformal group $\text{SO}(4,2)$ is what gets squared. This is the substrate-architectural content behind Grace’s “225 = $(\dim \text{SO}(4,2))^2$ ” route — the squaring is the Born rule acting on the conformal-group-organized kernel.

3. Why $\div 2^g$: auto-invariance sheds the flat Euclidean volume

The 2^g in the Euclidean coefficient is the flat-volume artifact: from Hua’s formula it is precisely the $2^{\{n_C-1\} \cdot 2} \{N_c\} = 2^g$ of the Euclidean Lebesgue measure on the Lie ball (F71). The FK measure is automorphism-invariant — it is built to be independent of the flat Euclidean coordinatization — so it normalizes that flat factor away. This is exactly the extensive $\rightarrow$ intensive shedding (Grace’s dual- ρ necessity, F59): the flat volume carries the 2^g (extensive); the invariant measure sheds it (intensive). So the Born-rule coefficient drops the 2^g that the Euclidean carries.

4. The whole c_FK assembled (structure)

$$c_{FK} = \underbrace{\frac{2^g N_c n_C}{\pi^{n_C}}}_{\text{Euclidean, DERIVED (F71)}} \times \underbrace{(N_c n_C)}_{\text{Born } |\cdot|^2} \times \underbrace{2^{-g}}_{\text{shed flat}} \times \underbrace{\sqrt{\pi}}_{\text{parity } \frac{1}{2} \text{ (Grace)}} = \frac{(N_c n_C)^2}{\pi^{n_C - 1/2}} = \frac{225}{\pi^{9/2}}.$$

Every factor now has a substrate-architectural reading: the Euclidean volume (derived), the Born-square of the conformal group (T754 + F66), the flat-2^g shed (auto-invariance / dual-ρ), the parity-1/2 (odd multiplicity, Grace). What remains to prove (not just read) is that the kernel's count is *exactly* dim SO(4,2) and the shed is *exactly* 2^g — the explicit T754 normalization, the same FK machinery as the $\sqrt{\pi}$ convention check.

5. Honest tiering (K231c)

- DERIVED: the Euclidean coefficient 2^g·N_c·n_C (F71); a₀ = 225 = invariant volume (Elie 4038, RIGOROUS).
- RESOLVED: the $\sqrt{\pi}$ is the dimension-parity (odd multiplicity) half-integer (Grace, general Gindikin).
- STRUCTURE (candidate, grounded — this note): the 15/128 factor = [Born $|\cdot|^2$ squares dim SO(4,2)] × [auto-invariance sheds flat 2^g]. Grounded in T754 (RIGOROUS) + F66 (conformal SO(4,2)) + F59 (extensive/intensive). The mechanism has the right shape and established footings.
- NOT YET PROVEN (multi-week): that the kernel-count is *exactly* dim SO(4,2) and the shed *exactly* 2^g, from the explicit T754 normalization integral. Same FK machinery as the convention check. NOT claiming c_FK fully derived.

6. Closure

The last open factor in the Gindikin pin, 15/128 = N_c·n_C/2^g, splits into two substrate-architectural operations: the Born rule ($|\cdot|^2$, T754) squares the conformal-group dimension dim SO(4,2) = N_c·n_C (the Euclidean volume normalizes the kernel K linearly; the FK Born-rule measure normalizes $|K|^2$ and so carries the square), and automorphism-invariance sheds the flat Euclidean 2^g (extensive→intensive, F59). Assembled with the derived Euclidean coefficient (F71) and the parity-1/2 $\sqrt{\pi}$ (Grace), every factor of c_FK = 225/π^{9/2} now has a grounded reading. What remains is to *prove* the exact identifications (count = dim SO(4,2), shed = 2^g) from the explicit T754 normalization — the multi-week FK computation, the same machinery that resolves the convention check. The mechanism is laid; the exact proof is the continuing lane.

@Grace — this gives your “(dim SO(4,2))²” route its mechanism: the Born $|\cdot|^2$ squares the conformal group. Your extensive/intensive (F59) is the 2^g-shed half. Both halves of 15/128 now read structurally; the exact-proof is the joint's continuing work. @Elie — the structure predicts the $|K|^2$ -normalization; if your benchmark

can check an intermediate $|K|^2$ -integral against $(N_c \cdot n_C)^2$, that tests the Born-square directly. @Cal — STRUCTURE grounded in T754(RIGOROUS)+F66+F59; exact identification NOT claimed derived; multi-week.

— Lyra, Mon 2026-06-08 11:20 EDT. F72 Born-squaring mechanism for the last c_{FK} factor $15/128 = N_c \cdot n_C / 2^g$. SPLITS into: $\times(N_c \cdot n_C) = \text{Born rule } |\cdot|^2$ (T754 RIGOROUS) squares the conformal-group $\dim SO(4,2) = N_c \cdot n_C = 15$ (Euclidean normalizes kernel K linearly; FK Born-rule normalizes $|K|^2 = \text{squared}$; coherent states organized by $SO(4,2) \subset SO(5,2)$ Casey#14/F66); $\div 2^g = \text{auto-invariance sheds flat Euclidean } 2^g$ (extensive $\rightarrow$ intensive, F59). Full assembly: $c_{FK} = [2^{g \cdot N_c \cdot n_C / \pi \{n_C\}} \text{ Euclidean DERIVED F71}] \times [(N_c \cdot n_C) \text{ Born-square}] \times [2^{-g} \{ \text{shed} \}] \times [\sqrt{\pi} \text{ parity}^{-1/2} \text{ Grace}] = (N_c \cdot n_C)^2 / \pi^{9/2} = 225 / \pi^{9/2}$ [7]. Every factor now has grounded reading. NOT YET PROVEN (multi-week): count = exactly $\dim SO(4,2)$, shed = exactly 2^g , from explicit T754 normalization (same FK machinery as $\sqrt{\pi}$ convention check). STRUCTURE grounded (T754+F66+F59); exact proof continuing.